

Md Tanvir Rouf Shawon

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 shawon-tanvir.github.io/  Scholar Profile

Profile

2nd-year PhD student in Computer Science at George Mason University specializing in Large Language Models, medical NLP, and explainable AI. Experienced in multi-agent simulation frameworks and evaluation systems for healthcare AI. Passionate about building safe and reliable AI for real clinical use.

Research Interests

Natural Language Processing, Health Informatics, Large Language Models, Machine Learning Application, Explainable Artificial Intelligence.

Skills

- **Programming Language:** *Python, Java, JavaScript, C, C++, PHP, HTML, CSS*
- **ML Tools:** PyTorch, Keras, Tensorflow, Scikit-learn, GANs, RAG
- **NLP:** Hugging Face, LangGraph, LangChain, GPT; **Visualization:** Matplotlib, Seaborn
- **Database:** MS SQL, MySQL, SQLite, Oracle PL SQL

Education

George Mason University
Ph.D. in Computer and Information Sciences

Fairfax, Virginia, USA
August 2024 - Current

Ahsanullah University of Science and Technology
B. Sc. in Computer Science and Engineering

Dhaka, Bangladesh
2016 - 2021

Notre Dame College, Dhaka
Higher Secondary Certificate

Dhaka, Bangladesh
2013 - 2015

Selected Publications

(* denotes equal contribution)

- **Md. Tanvir Rouf Shawon**, G. M. Shahariar Shibli, Farzad Ahmed & Sajib Kumar Saha Joy "**Explainable cost-sensitive deep neural networks for brain tumor detection from brain MRI images considering data imbalance**" in *Multimedia Tools and Applications, Springer*, 2025
- G. M. Shahariar Shibli*, **Md. Tanvir Rouf Shawon***, Faisal Muhammad Shah, Mohammad Shafiul Alam & Md. Shahriar Mahbub "**Bengali Fake Reviews: A Benchmark Dataset and Detection System**" in *Neurocomputing, Elsevier*, 2024
- Kazi Elahi, Tasnuva Rahman, Shakil Shahriar, Samir Sarker, **Md. Tanvir Rouf Shawon**, and G. M. Shibli "**A Comparative Analysis of Noise Reduction Methods in Sentiment Analysis on Noisy Bangla Texts**" in *9th Workshop on Noisy and User-generated Text (W-NUT 2024), EACL*, 2024
- G. M. Shahariar Shibli, **Md. Tanvir Rouf Shawon**, Anik Hassan Nibir, Md. Zayed Miandad & Nibir Chandra Mandal "**Automatic back transliteration of Romanized Bengali (Banglish) to Bengali**" in *Iran Journal of Computer Science, Springer*, 2023
- **Md. Tanvir Rouf Shawon**, G. M. Shahariar Shibli, Faisal Muhammad Shah, Mohammad Shafiul Alam & Md. Shahriar Mahbub "**Bengali Fake Review Detection using Semi-supervised Generative Adversarial Networks**" in *5th International Conference on Natural Language Processing (ICNLP), Guangzhou, China, IEEE*, 2023
- **Md Tanvir Rouf Shawon***, Raihan Tanvir* & Md. Golam Rabiul Alam "**Bengali Handwritten Digit Recognition using CNN with Explainable AI**" in *4th International Conference on Sustainable Technologies for Industry 4.0 (STI), Dhaka, Bangladesh, IEEE*, 2022

- Raihan Tanvir*, **Md Tanvir Rouf Shawon***, Md Humaion Kabir Mehedi, Md Motahar Mahtab & Annajiat Alim Rasel "A GAN-BERT Based Approach for Bengali Text Classification with a Few Labeled Examples" in *Distributed Computing and Artificial Intelligence, 19th International Conference, L'Aquila, Italy, Springer, 2022*
- **MD Tanvir Rouf Shawon***, Raihan Tanvir*, Humaira Ferdous Shifa, Susmoy Kar & Mohammad Imrul Jubair "Jamdani Motif Generation using Conditional GAN" in *23rd International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, IEEE, 2020*

Research Grant

(CO-PI) Bengali Fake Review: A Benchmark Dataset and Detection System. 2022
Internal Research Grant supported by Committee for Advance Studies and Research (CASR), AUST

Conference Presentation

- Bengali Fake Review Detection using Semi-supervised Generative Adversarial Networks, *5th International Conference on Natural Language Processing (ICNLP)*, Guangzhou, China. [Presentation link: <https://youtu.be/rd5N6z5-HOU>] March 24-26, 2023
- Effectiveness of Transformer Models on IoT Security Detection in StackOverflow Discussions, *1st International Conference on Information and Communication Technology for Development*, Khulna, Bangladesh. [Presentation link: <https://youtu.be/Y4ntugjHEGM>] July 29-30, 2022

Teaching Experience

Ahsanullah University of Science and Technology (AUST)

Lecturer

June 2021 - July 2024

Department of Computer Science & Engineering

Courses: Soft Computing, Pattern Recognition, Algorithms, Object Oriented Programming

Awards & Achievements

- Dean's List of Honor, AUST 2021
- Government Scholarship, Sylhet 2010, 2013
- I-Genius, Grameenphone Prothom Alo Internet Festival 2013

Professional Activities

- Co-opt member (Extra Curricular Activities) - AUST CSE SOCIETY
- Coach, Aust.0Degree & AUST Samakal: ICPC Asia Dhaka Regional Onsite Contest
- Advisor of Cefalo CodeFiesta 2022: AUST Inter-University Programming Contest
- Organizer of Intra-AUST Programming Contest Spring 2019